Proposed Solution

DATE	02 JULY 2026
TEAM ID	
PROJECT NAME	**A Comprehensive Measure of Well-Being: Human Development Index (HDI) Predictor**
MAXIMUM MARKS	2 MARKS

1. Executive Summary

This document describes the design and implementation details of the Human Development Index (HDI) Prediction System. The solution combines a machine learning model, a Flask web server, and a responsive frontend to create an interactive tool for predicting and visualizing human development levels.

---

2. Technical System Modules

2.1 Preprocessing Module (`data_preprocessing.py`)

- **Data Loading**: Loads the HDI dataset (`data/hdi_data.csv`) using Pandas.
- **Feature Selection**: Selects features: `Country`, `Life_Expectancy`, `Expected_Years_Schooling`, `Mean_Years_Schooling`, `GNI_Per_Capita`.
- **Imputation**: Cleans missing data using mean imputation for numeric indicators.
- **Categorical Encoding**: Fits a `LabelEncoder` to convert the `Country` category column to numeric values.
- **Data Partitioning**: Splits the data into 80% training and 20% testing sets.

2.2 Machine Learning Module (`ml_model.py`)

- **Algorithm**: Linear Regression.
- **Model Object**: `sklearn.linear_model.LinearRegression`.
- **Metrics**: Evaluates model performance using Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2).
- **Serialization**: Saves the trained model and label encoder as pickle files:
- `models/hdi_model.pkl`
- `models/encoder.pkl`

2.3 Application Server Module (`app.py`)

- **Routing**: Implements endpoint routers using Flask:
- `/` (GET) renders the landing page dashboard.
- `/predict` (GET) renders the parameter form.
- `/predict` (POST) processes inputs, generates predictions, and displays results.
- **Post-Processing**: Clamps the predicted score to `[0.0, 1.0]` to handle extreme inputs.
- **Tier Classification**: Classifies predicted scores into UNDP categories (Very High, High, Medium, Low).
- **Insights Engine**: Matches the predicted tier to targeted recommendations.

2.4 User Interface Layer (`templates/` & `static/`)

- **Design Theme**: Dark-mode dashboard using modern glassmorphism (frosted glass) cards.
- **Visual Features**:
- Color-coded badges for development tiers.
- Custom progress bar that acts as a visual progress gauge for the predicted score.
- Actionable recommendations displayed in bulleted format.
- **Responsive Layout**: Designed using CSS Flexbox and Grid to ensure responsiveness across desktop, tablet, and mobile browsers.

---

3. Model Parameters and Coefficients

The Linear Regression model evaluates features according to the following weights:

Feature Name	Coefficient Value	Impact on HDI Score
Expected Years of Schooling (EYS)	`0.0128`	High Impact
Mean Years of Schooling (MYS)	`0.0107`	High Impact
Life Expectancy	`0.0057`	Moderate Impact
GNI per Capita	`< 0.0001`	Low Direct Impact
Country (encoded)	`< 0.0001`	Low Direct Impact

Interpretation

- **Education Indicators** (EYS and MYS) have the highest direct impact on the predicted HDI score.
- **Life Expectancy** has a moderate direct impact.
- **GNI per Capita** has a low direct coefficient because the model fits the raw indicator. In the actual index, GNI is log-transformed to account for diminishing returns on income.